Data Low QC

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Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(writexl))
suppressMessages(library(SingleR))
suppressMessages(library(scuttle))
suppressMessages(library(scRNAseq))
suppressMessages(library(RColorBrewer))
suppressMessages(library(stringr))
suppressMessages(library(formatR))
```

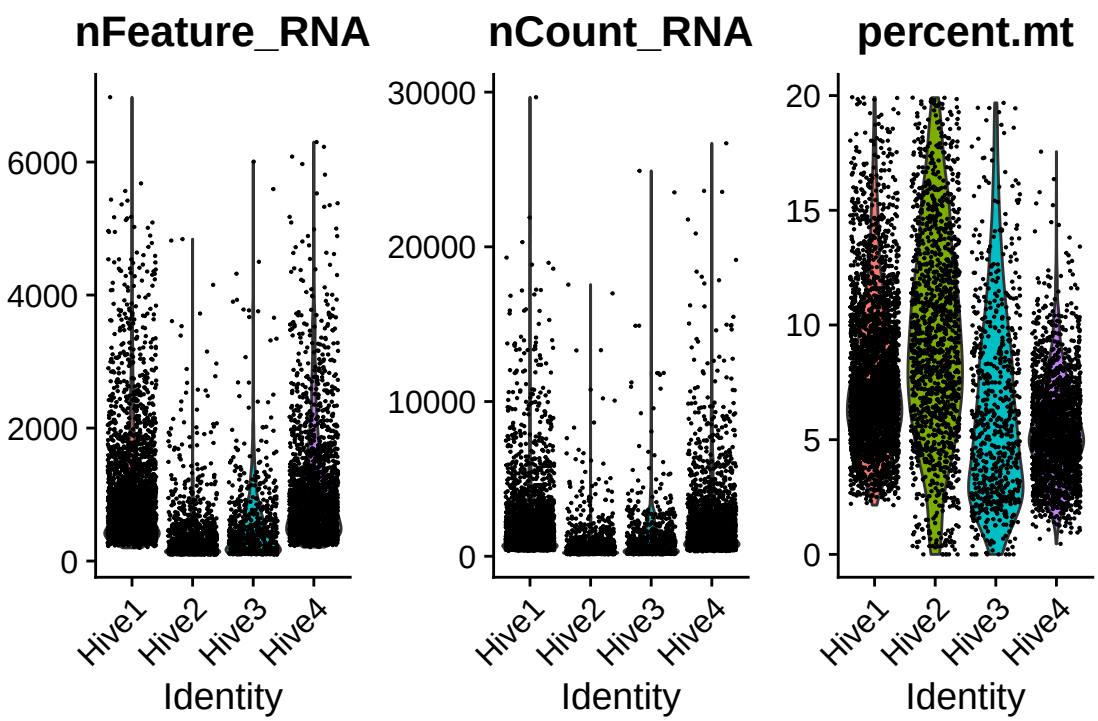
Loading data

```
all_dirs <- dir(path = "./Counts", full.names = T)

list_sample <- list()
for (dir in all_dirs) {

  files <- list.files(dir)
  Count_file <- files[grep("TCM.tsv.gz$", files)]
  Countdata <- read.table(paste0(dir,"/",Count_file), sep = "\t", header = T,
row.names=1)
```

```
  Lavage_cellsHive <- CreateSeuratObject(counts = Countdata, project =
str_sub(dir,-5,-1), min.features = 100)
  list_sample <- append(list_sample, Lavage_cellsHive)

}

Lavage_cellsHive <- merge(list_sample[[1]], y = list_sample[-1], add.cell.ids =
c("1","2","3", "4"), project = "bronchoalveolar_cells")
```

QC

```
Lavage_cellsHive[["percent.mt"]] <- PercentageFeatureSet(Lavage_cellsHive, pattern =
"^MT-")# MT : human cells
Lavage_cellsHive <- subset(Lavage_cellsHive, subset = nFeature_RNA < 8000 & percent.mt <
20)
VlnPlot(Lavage_cellsHive, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol
= 3, pt.size = 0.1, group.by = "orig.ident")
```




```
#ggsave("plot/QC.png", width = 12, height = 6)
```

Normalisation + scalin

```
Lavage_cellsHive <- NormalizeData(Lavage_cellsHive, normalization.method =
"LogNormalize", scale.factor = 10000)
Lavage_cellsHive <- FindVariableFeatures(Lavage_cellsHive, selection.method = "vst",
nfeatures = 2000)
```

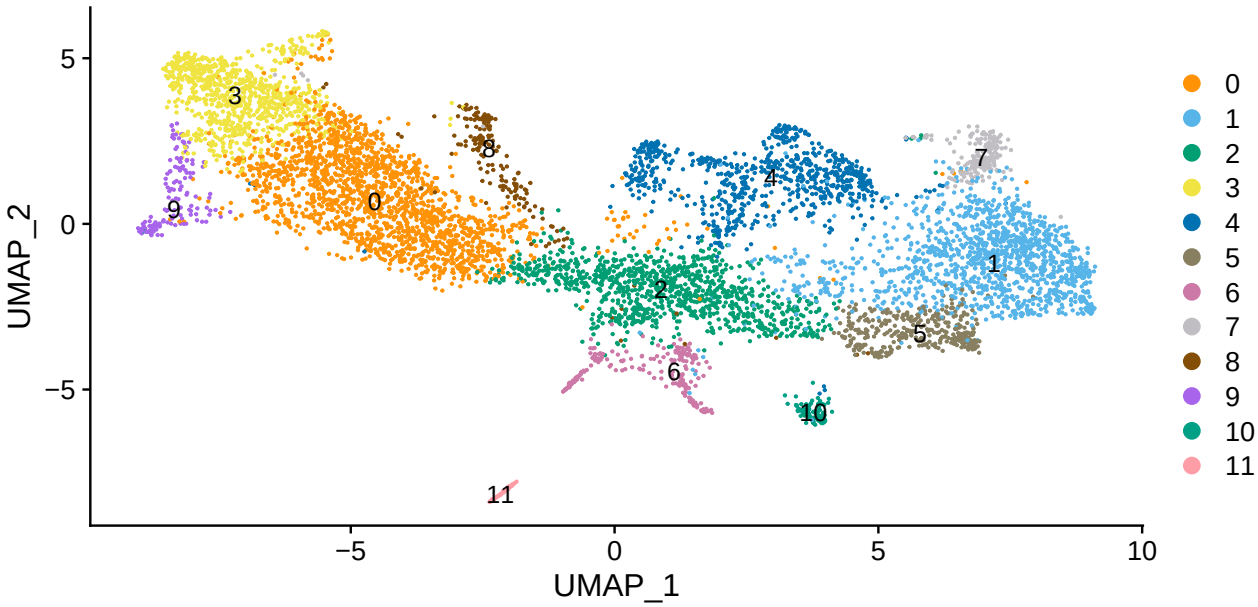
```
all.genes <- rownames(Lavage_cellsHive)
Lavage_cellsHive <- ScaleData(Lavage_cellsHive, features = all.genes)
```

Dimension reduction

```
Lavage_cellsHive <- RunPCA(Lavage_cellsHive, features = VariableFeatures(object =
Lavage_cellsHive))

Lavage_cellsHive <- FindNeighbors(Lavage_cellsHive, dims = 1:15)
Lavage_cellsHive <- FindClusters(Lavage_cellsHive, resolution = 0.5)
Lavage_cellsHive <- RunUMAP(Lavage_cellsHive, dims = 1:15)
```

umap

```
Lavage_cellsHive$Annotation <- Lavage_cellsHive$seurat_clusters
Lavage_cellsHive$Annotation <- as.factor(Lavage_cellsHive$Annotation)
levels(Lavage_cellsHive$Annotation) <- c("C1-AM", "C2-T cell", "C3-Unidentified",
"C4-AM", "C5-Neu", "C6-T cell", "C7-Mast Cell", "C8-Mo-Mac", "C9-AM", "C10-AM",
"C11-Eos", "C12-Epithelial Cell")


my_palette <- c("#ff9305", "#56B4E9", "#009E73", "#F0E442", "#0072B2",
                "#877f5f", "#CC79A7", "#c0bec2", "#854e07", "#a865eb",
                "#00A087", "#FF9DA7", "#A55AFF", "#8C4B4B", "#00A08E")

DimPlot(Lavage_cellsHive, reduction = "umap", label = T, cols = my_palette)
```

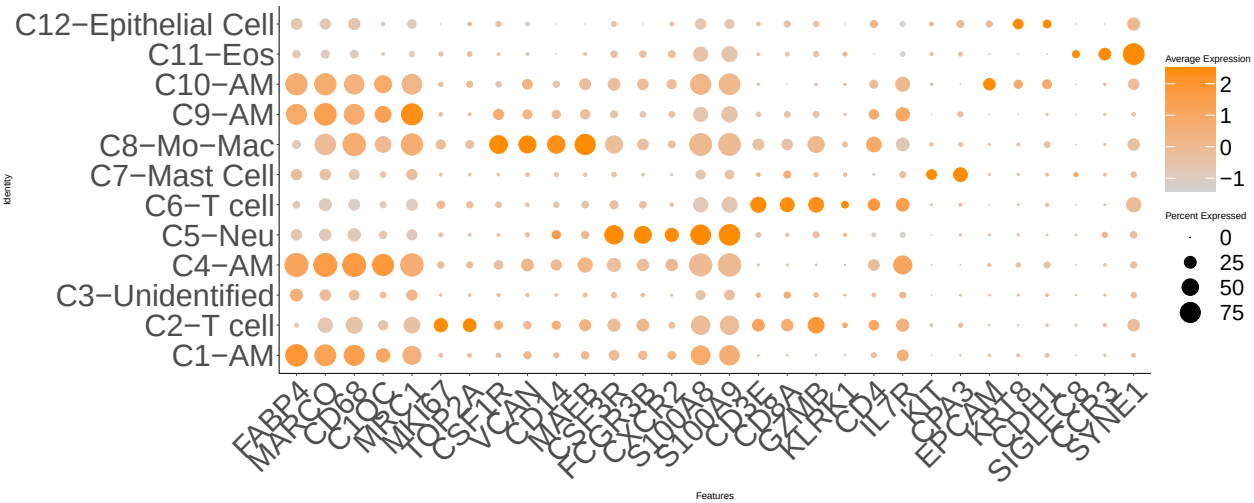



```
Marker_gene <- c("FABP4", "MARCO", "CD68", "C1QC", "MRC1", "MKI67", "TOP2A", "CSF1R",
"VCAN", "CD14",
"MAFB", "CSF3R", "FCGR3B", "CXCR2", "S100A8", "S100A9", "CD3E", "CD8A", "GZMB",
"KLGR1", "KLRK1", "CD4", "IL7R", "KIT", "CPA3", "EPCAM", "KRT8", "CDH1", "SIGLEC8",
"CCR3",
```

```
"SYNE1")

DotPlot(Lavage_cellsHive, features = Marker_gene, cols = c("lightgray", "darkorange"),
group.by = "Annotation") + theme_classic() + scale_size(range = c(0,12)) +
theme(axis.text.x = element_text(angle = 45, hjust = 1), axis.text = element_text(size =
35),
           legend.text   = element_text(size = 25), legend.key.height= unit(1, 'cm'),
        legend.key.width= unit(2, 'cm'))
```

```
## Warning in FetchData.Seurat(object = object, vars = features, cells = cells):
## The following requested variables were not found: KLGR1
```

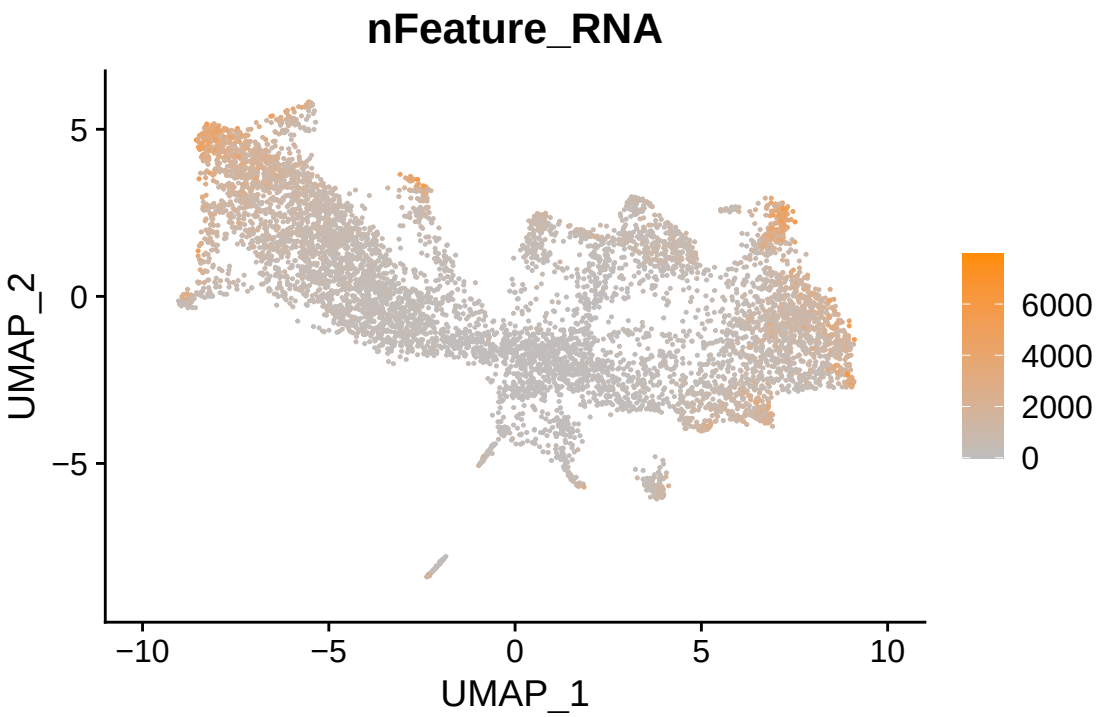
```
## Scale for size is already present.
## Adding another scale for size, which will replace the existing scale.
```




```
#ggsave("plot_lastMinute/Dotplot_Hive.pdf", width = 35, height = 10)
```

```
FeaturePlot(Lavage_cellsHive, features = c("nFeature_RNA"))+ scale_color_gradient(low =
"grey", high = "darkorange", limit = c(0,7960) )
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```




```
#ggsave("/nFeature_hive.pdf", width = 8, height = 4)
```

```
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] formatR_1.14                stringr_1.5.0
##  [3] RColorBrewer_1.1-3          scRNAseq_2.14.0
##  [5] scuttle_1.10.1              SingleCellExperiment_1.22.0
```

```
##  [7] SingleR_2.2.0               SummarizedExperiment_1.30.2
##  [9] Biobase_2.60.0              GenomicRanges_1.52.0
## [11] GenomeInfoDb_1.36.0         IRanges_2.34.0
## [13] S4Vectors_0.38.1            BiocGenerics_0.46.0
## [15] MatrixGenerics_1.12.2       matrixStats_1.0.0
## [17] writexl_1.4.2               ggplot2_3.4.2
## [19] patchwork_1.1.2             SeuratObject_4.1.3
## [21] Seurat_4.3.0                dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21              splines_4.3.3
##   [3] later_1.3.1                   BiocIO_1.10.0
##   [5] filelock_1.0.2                bitops_1.0-7
##   [7] tibble_3.2.1                  polyclip_1.10-4
##   [9] XML_3.99-0.14                 lifecycle_1.0.3
##  [11] ensembldb_2.24.0              globals_0.16.2
##  [13] lattice_0.22-5                MASS_7.3-60.0.1
##  [15] magrittr_2.0.3                plotly_4.10.2
##  [17] rmarkdown_2.23                yaml_2.3.7
##  [19] httpuv_1.6.11                 sctransform_0.3.5
##  [21] spam_2.9-1                    sp_2.0-0
##  [23] spatstat.sparse_3.0-2         reticulate_1.30
##  [25] cowplot_1.1.1                 pbapply_1.7-2
##  [27] DBI_1.1.3                     abind_1.4-5
##  [29] zlibbioc_1.46.0               Rtsne_0.16
##  [31] purrr_1.0.1                   AnnotationFilter_1.24.0
##  [33] RCurl_1.98-1.12               rappdirs_0.3.3
##  [35] GenomeInfoDbData_1.2.10       ggrepel_0.9.3
##  [37] irlba_2.3.5.1                 listenv_0.9.0
##  [39] spatstat.utils_3.0-3          goftest_1.2-3
##  [41] spatstat.random_3.1-5         fitdistrplus_1.1-11
##  [43] parallelly_1.36.0             DelayedMatrixStats_1.22.1
##  [45] leiden_0.4.3                  codetools_0.2-19
##  [47] DelayedArray_0.26.3           xml2_1.3.4
##  [49] tidyselect_1.2.0              farver_2.1.1
##  [51] ScaledMatrix_1.8.1            BiocFileCache_2.8.0
##  [53] spatstat.explore_3.2-1        GenomicAlignments_1.36.0
##  [55] jsonlite_1.8.7                ellipsis_0.3.2
##  [57] progressr_0.13.0              ggridges_0.5.4
##  [59] survival_3.5-8                progress_1.2.2
##  [61] tools_4.3.3                   ica_1.0-3
##  [63] Rcpp_1.0.11                   glue_1.6.2
##  [65] gridExtra_2.3                 xfun_0.39
##  [67] withr_2.5.0                   BiocManager_1.30.21
##  [69] fastmap_1.1.1                 fansi_1.0.4
##  [71] digest_0.6.33                 rsvd_1.0.5
##  [73] R6_2.5.1                      mime_0.12
##  [75] colorspace_2.1-0              scattermore_1.2
##  [77] tensor_1.5                    biomaRt_2.56.1
##  [79] spatstat.data_3.0-1           RSQLite_2.3.1
##  [81] utf8_1.2.3                    tidyr_1.3.0
##  [83] generics_0.1.3                data.table_1.14.8
##  [85] rtracklayer_1.60.0            prettyunits_1.1.1
##  [87] httr_1.4.6                    htmlwidgets_1.6.2
```

```
##  [89] S4Arrays_1.2.1                uwot_0.1.16
##  [91] pkgconfig_2.0.3               gtable_0.3.3
##  [93] blob_1.2.4                    lmtest_0.9-40
##  [95] XVector_0.40.0                htmltools_0.5.5
##  [97] dotCall64_1.0-2               ProtGenerics_1.32.0
##  [99] scales_1.2.1                  png_0.1-8
## [101] knitr_1.43                    rstudioapi_0.14
## [103] rjson_0.2.21                  reshape2_1.4.4
## [105] curl_5.0.1                    nlme_3.1-164
## [107] zoo_1.8-12                    cachem_1.0.8
## [109] BiocVersion_3.17.1            KernSmooth_2.23-22
## [111] vipor_0.4.5                   parallel_4.3.3
## [113] miniUI_0.1.1.1                AnnotationDbi_1.62.1
## [115] ggrastr_1.0.2                 restfulr_0.0.15
## [117] pillar_1.9.0                  grid_4.3.3
## [119] vctrs_0.6.3                   RANN_2.6.1
## [121] promises_1.2.0.1              BiocSingular_1.16.0
## [123] dbplyr_2.3.2                  beachmat_2.16.0
## [125] xtable_1.8-4                  cluster_2.1.6
## [127] beeswarm_0.4.0                evaluate_0.21
## [129] GenomicFeatures_1.52.0        Rsamtools_2.16.0
## [131] cli_3.6.1                     compiler_4.3.3
## [133] rlang_1.1.1                   crayon_1.5.2
## [135] future.apply_1.11.0           labeling_0.4.2
## [137] ggbeeswarm_0.7.2              plyr_1.8.8
## [139] stringi_1.7.12                viridisLite_0.4.2
## [141] deldir_1.0-9                  BiocParallel_1.34.2
## [143] munsell_0.5.0                 Biostrings_2.68.1
## [145] lazyeval_0.2.2                spatstat.geom_3.2-4
## [147] Matrix_1.6-1                  ExperimentHub_2.8.0
## [149] hms_1.1.3                     sparseMatrixStats_1.12.0
## [151] bit64_4.0.5                   future_1.33.0
## [153] KEGGREST_1.40.0               shiny_1.7.4.1
## [155] highr_0.10                    interactiveDisplayBase_1.38.0
## [157] AnnotationHub_3.8.0           ROCR_1.0-11
## [159] igraph_1.5.0.1                memoise_2.0.1
## [161] bit_4.0.5
```